DO YOU KNOW?

A linear differential equation is always of first degree but every differfential equation of the first degree need not be linear

$$\frac{d^2y}{dx^2}-\frac{3dy}{dx}+2y^2=0$$

is not linear.

Example 4.21

Solve $\cos^2 x\frac{dy}{dx}+y=\tan x$

Solution:

The given equation can be written as

$$\frac{dy}{dx}+\frac{1}{\cos^2 x}y=\frac{\tan x}{\cos^2 x}$$

$$\frac{dy}{dx}+y\sec^2 x = \tan x\sec^2 x$$

It is of the form $\frac{dy}{dx}+Py=\text{Q}$

Here $\text{P}=\sec^2 x, Q = \tan x\sec^2 x$

$$\int Pdx = \int \sec^2 x\,dx=\tan x$$

$$\text{I.F} = e^{\int pdx}=e^{\tan x}$$

The required solution is

$$y(I.F) = \int Q(I.F)dx+c$$

$$ye^{\tan x} = \int \tan x\sec^2 x e^{\tan x}\,dx+c$$

Put tan x = t

Then $\sec^2 x dx =$ dt

$$\therefore ye^{\tan x} = \int te^t\,dt+c$$

$$= \int td(e^t)+c$$

$$= te^t-e^t+c$$

$$= \tan x\,e^{\tan x}-e^{\tan x}+c$$

$$ye^{\tan x} = e^{\tan x}(\tan x-1)+c$$

Example 4.22

Solve $(x^2+1)\frac{dy}{dx}+2xy=4x^2$

Solution:

The given equation can be reduced to

$$\frac{dy}{dx}+\frac{2x}{x^2+1}y = \frac{4x^2}{x^2+1}$$

It is of the form $\frac{dy}{dx}+Py = \text{Q}$

Here $P=\frac{2x}{x^2+1}, Q=\frac{4x^2}{x^2+1}$

$$\int Pdx=\int\frac{2x}{x^2+1}dx = \log(x^2+1)$$

$$\text{I.F} = e^{\int pdx}=e^{\log(x^2+1)}=x^2+1$$

The required solution is

$$y(IF) = \int Q(I.F)dx+c$$

$$y(x^2+1) = \int\frac{4x^2}{x^2+1}(x^2+1)dx+c$$

$$y(x^2+1) = \frac{4x^3}{3}+c$$

Example 4.23

Solve $\frac{dy}{dx}-3y\cot x=\sin 2x$ given that $y=2$ when $x=\frac{\pi}{2}$

Solution:

Given $\frac{dy}{dx}-(3\cot x).y=\sin 2x$

It is of the form $\frac{dy}{dx}+Py=\text{Q}$

Here $P=-3\cot x, Q=\sin 2x$

XII Std - Business Maths & Stat EM Chapter 4.indd 93 2/28/2020 6:02:38 PM